

Enterprise AIOps Platform - Complete Master Architecture

Target Platform: Google Cloud Platform (GCP) System of Record: ServiceNow ITSM Version: 2.0 (Production Blueprint) Generated: August 2026

Enterprise AIOps Platform - Master Architecture Blueprint

⚠ IMPORTANT

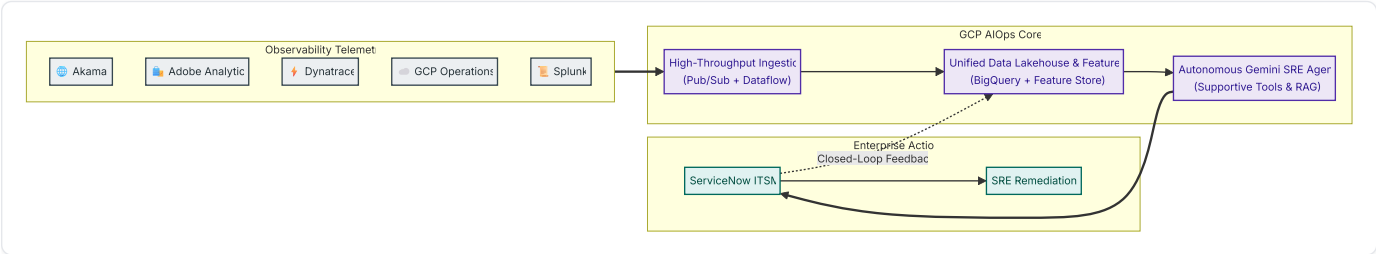
> This comprehensive document consolidates all 8 architectural domain specifications for the Enterprise AIOps Platform, covering Ingestion, Canonical Data Contracts, Analytical Lakehouse, AI Feature Store, Autonomous Gemini SRE Agent, and ServiceNow ITSM Integration.

Enterprise AIOps Platform - High-Level Architecture Overview

1. Executive Summary & Business Objective

Modern global enterprise operations face extreme telemetry fragmentation. For a large multi-channel retail organization, operations span thousands of edge locations, extensive cloud footprints, and millions of customer transactions per hour.

The **Enterprise AIOps Platform** consolidates disparate observability streams from the SRE team's 5 core tools—**Google Cloud Platform (GCP)**, **Dynatrace**, **Akamai**, **Splunk**, and **Adobe Analytics**—into a centralized intelligence engine natively hosted on Google Cloud Platform. The platform correlates business transactions with technical performance to automate incident triage, accelerate root cause identification, and trigger proactive remediation in **ServiceNow**.



2. Core Architectural Pillars

2.1 Multi-Source Ingestion to GCP Core

While telemetry originates from diverse specialized tools across edge networks, cloud platforms, and enterprise data centers, **all centralized stream processing, analytical storage, and AI reasoning workloads run natively on Google Cloud Platform (GCP)**. This prevents vendor lock-in across tool providers while consolidating data for cross-domain ML models.

2.2 ServiceNow as the Definitive ITSM System of Record

ServiceNow is the enterprise system of record for all incident lifecycles, configuration item (CMDB) topologies, on-call schedules, and automated remediation audit logs. The AIOps layer communicates bidirectionally via standard ServiceNow REST APIs to enrich existing records, generate consolidated incidents, and bypass manual L1 triage.

2.3 Event-Driven Streaming & Elastic Scale

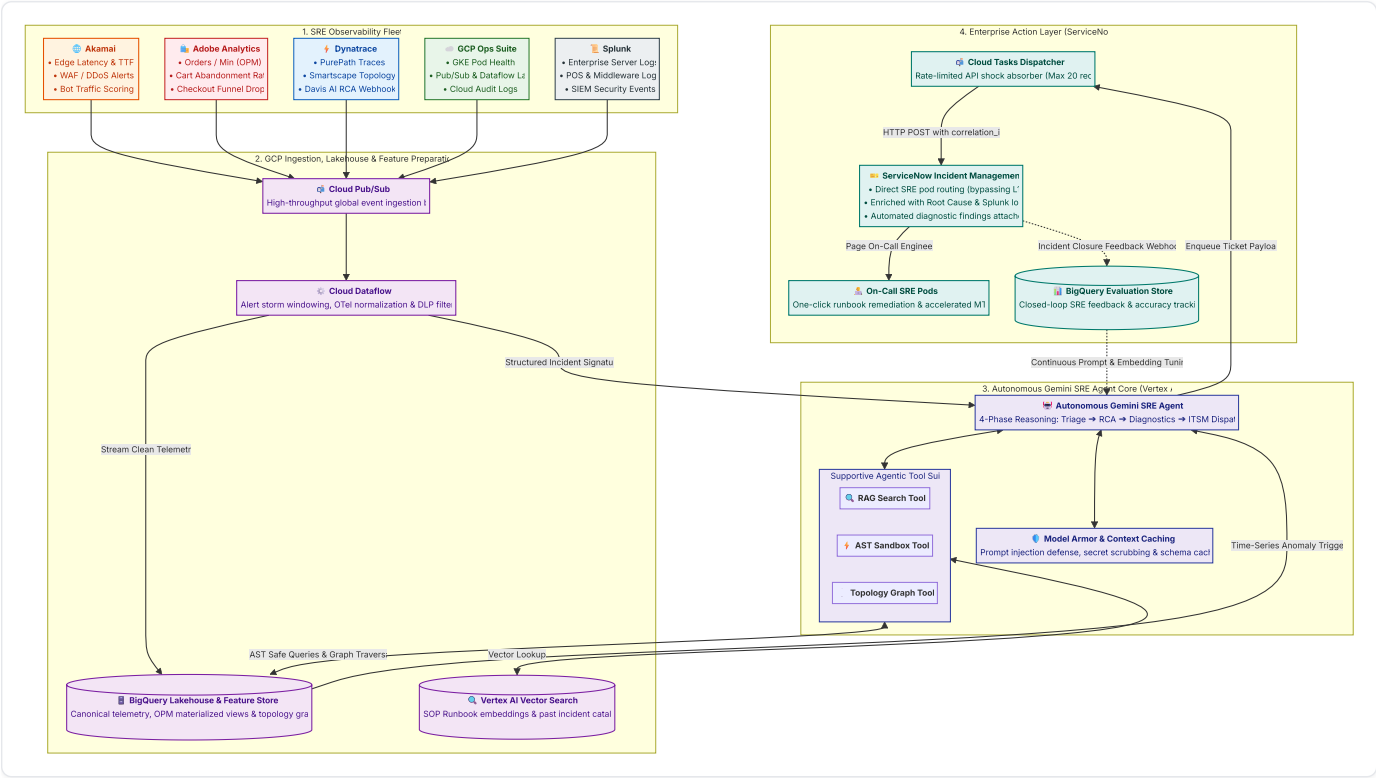
The platform is built on an event-driven architecture using **Cloud Pub/Sub** and **Cloud Dataflow (Apache Beam)**. It comfortably sustains normal loads of 150,000+ Events Per Second (EPS) and auto-scales to absorb peak promotional traffic (e.g., Black Friday / Cyber Monday surges exceeding 500,000 EPS) without message loss or latency degradation.

2.4 Actionable AI & Autonomous Incident Diagnostics

Rather than producing passive dashboards, the AI architecture utilizes the **Autonomous Gemini SRE Agent** with **Supportive Function Calling Tools** to execute tangible operational workflows:

- De-duplicating cascading alerts into unified incident signatures.
- Retrieving matching Standard Operating Procedures (SOPs) from **Vertex AI Vector Search**.
- Automatically executing non-destructive diagnostic queries and posting synthesized findings to ServiceNow tickets.
- Capturing closed-loop SRE feedback upon ticket closure to continuously evaluate and tune prompts.

3. End-to-End Architectural Flow



4. Architectural Domain Map & Detailed References

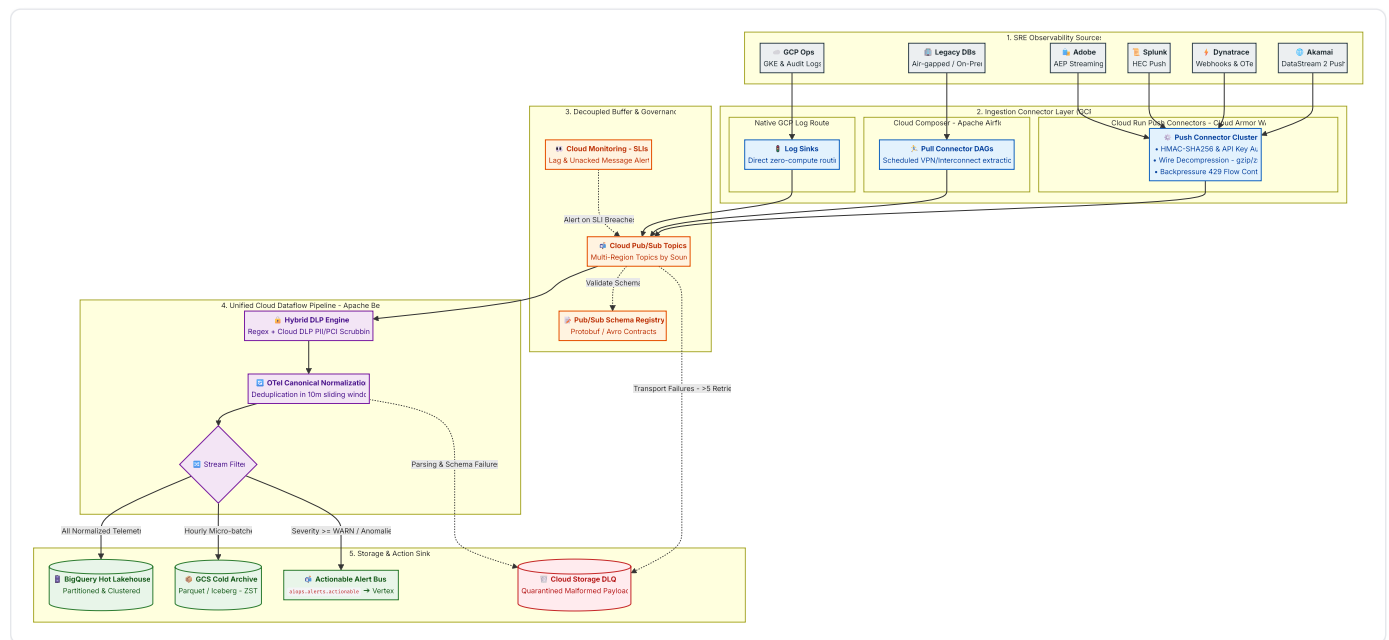
Architecture Domain	Scope & Responsibilities	Deep-Dive Specification
01. Ingestion Architecture	Transport protocols, Pub/Sub topologies, Dataflow Beam pipelines, Cloud DLP scrubbing, dead-letter queues, and unified telemetry matrix for all 5 observability tools.	01_ingestion/README.md source telemetry matrix.md
02. Storage, Lakehouse & Data Processing	BigQuery dataset designs, table partitioning by ingestion time, Parquet on GCS cold storage tiering, stateful alert clustering, time-series resampling, and topology graph ETL.	lakehouse architecture.md data_processing_and_feature_store.md
03. Intelligence & Reasoning	Autonomous Gemini SRE Agent runtime, 4-phase reasoning lifecycle, supportive function calling tools (RAG, AST Sandbox, Topology Graph), Model Armor guardrails, and context caching.	aiops_intelligence_layer.md
04. ITSM & Remediation	ServiceNow Incident Management integration, bidirectional lifecycle synchronization, CMDB topological sync, and automated diagnostic execution.	servicenow_integration.md

Ingestion Layer & Streaming Pipelines Architecture

This document outlines the **Enterprise AIOps Ingestion Engine**, which captures high-throughput telemetry from multi-cloud and edge observability sources, standardizes and sanitizes it in real time, and routes it to downstream analytics and incident response systems natively on **Google Cloud Platform (GCP)**.

1. High-Level Ingestion Architecture

The ingestion architecture follows a decoupled, connector-based pattern capable of handling 150,000 to 500,000+ Events Per Second (EPS) with sub-second latency and zero data loss.



2. The Ingestion Connector Layer

Telemetry sources exhibit different networking capabilities and security constraints. We utilize three distinct connector patterns to ingest telemetry securely into GCP:

2.1 Push Connectors (Modern SaaS & Edge Ingress)

Used by sources capable of real-time outbound streaming (**Akamai DataStream 2**, **Dynatrace Webhooks**, **Splunk HEC**, **Adobe Analytics AEP**).

- Ingress Security:** Traffic passes through **Cloud Armor WAF** with strict IP allowlisting (filtering for vendor CIDR ranges) and volumetric DDoS rate limiting.
- Compute Layer:** Auto-scaling, stateless **Cloud Run** container services deployed across multiple zones.
- Authentication & Webhook Validation:**
 - API Keys & Tokens:** Verified dynamically against **GCP Secret Manager** with in-memory caching.
 - HMAC-SHA256 Signatures:** Validates webhook payload authenticity using vendor signature headers (e.g., `X-Akamai-Signature`, `X-Adobe-Signature`) to prevent request spoofing.
- Wire Compression:** Enforces `Content-Encoding: gzip / zstd` on inbound streams, reducing cross-cloud data transfer volume and egress bandwidth costs by up to 70%.
- Backpressure & Flow Control:** If downstream Pub/Sub publishing encounters transient backpressure, Cloud Run returns `HTTP 429 (Too Many Requests)` or `HTTP 503` with a `Retry-After` header, prompting upstream SaaS forwarders to buffer and retry exponentially.

2.2 Pull Connectors (Legacy & Air-Gapped Systems)

Used by on-premise relational databases, mainframes, or network appliances that cannot push outbound traffic.

- Orchestration:** **Cloud Composer (Managed Apache Airflow)** schedules and executes extraction DAGs.
- Networking:** Traffic travels over private, encrypted tunnels via **Cloud VPN** or **Cloud Interconnect**.
- Extraction & Retries:** Airflow operators extract delta records, convert them to canonical JSON/Avro, and publish to Cloud Pub/Sub with automatic exponential backoff retries.

2.3 Native Connectors (GCP Operations Suite)

Used natively by Google Cloud Platform workloads (**Google Kubernetes Engine (GKE)**, **Cloud Run**, **Cloud Audit Logs**).

- **Mechanism:** Leverages **Cloud Logging Log Router Sinks** pointing directly to Cloud Pub/Sub topics.
- **Advantage:** Zero-compute, zero-maintenance ingestion with sub-second delivery latency and zero egress cost.

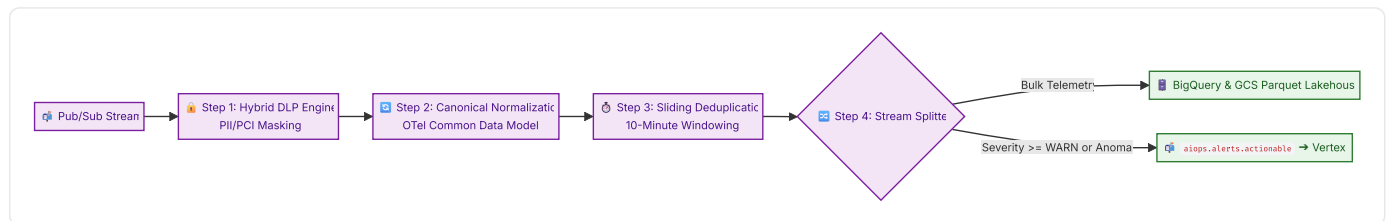
3. Decoupled Buffer & Stream Processing

3.1 Cloud Pub/Sub (The Multi-Region Event Bus)

- **Topic Isolation:** Dedicated, isolated topics per source (e.g., `telemetry.akamai.raw`, `telemetry.dynatrace.raw`, `telemetry.splunk.raw`) prevent single-source traffic storms (such as a DDoS attack logged by Akamai) from starving other telemetry streams.
- **Multi-Region Durability:** Topics are configured with multi-region message routing policies to ensure continuous ingestion availability during single-region GCP outages.

3.2 Cloud Dataflow (Unified Apache Beam Streaming Pipeline)

A unified, horizontally auto-scaling streaming pipeline processes incoming telemetry in four sequential stages:



1. **Hybrid DLP Engine:** Fast, in-memory regex tokenizers redact high-risk payment card numbers (PCI-DSS) and social security numbers. Unstructured log bodies are asynchronously scrubbed using the **Cloud Data Loss Prevention (DLP) API**.
2. **Canonical Normalization:** Standardizes disparate formats (Akamai JSON, Splunk HEC, Dynatrace PurePath, Adobe clickstream) into an **OpenTelemetry-aligned canonical schema** (`aiops_lakehouse.telemetry_canonical`).
3. **Sliding Window Deduplication:** Removes duplicate events within a 10-minute sliding window based on `(source_tool, source_event_id, timestamp)`.
4. **Intelligent Stream Splitting:**
 - **Bulk Analytical Stream:** Routes 100% of clean telemetry to BigQuery (Hot Tier) and GCS Parquet (Cold Tier).
 - **Actionable Alert Stream:** High-severity anomalies and warning events (`severity ≥ WARN` or Davis AI problems) are extracted and published to the high-priority Pub/Sub topic `aiops.alerts.actionable` for consumption by the **Vertex AI Semantic Router**.

4. Reliability, Robustness & Governance Patterns

4.1 Schema Management & Dynamic Evolution

- **Pub/Sub Schema Registry:** Enforces Protobuf and Avro schema contracts at topic ingress.
- **Schema Drift Protection:** Incoming records that introduce backward-compatible fields are parsed dynamically into `raw_attributes JSON`, preventing Dataflow worker crashes while preserving full fidelity.

4.2 Multi-Tier Dead-Letter Queues (DLQs)

To prevent unparseable or poisoned payloads from blocking the pipeline:

1. **Transport-Level DLQ (Pub/Sub):** If Dataflow fails to acknowledge a message after 5 delivery attempts, Pub/Sub diverts it to a native dead-letter topic (`telemetry.*.dlq`).
2. **Application-Level DLQ (Dataflow Side-Output):** Payloads that parse structurally but fail semantic validation (e.g., corrupted timestamps) are emitted via Beam side-outputs.
3. **Storage & Replay:** Both DLQ streams persist to a date-partitioned **Cloud Storage DLQ Bucket** (`gs://aiops-dlq-bucket/YYYY/MM/DD/`) for offline inspection, alerting, and automated replay.

4.3 Ingestion Service Level Indicators (SLIs) & Alerting

The platform team continuously monitors pipeline health in **Cloud Monitoring**:

Metric Name	Target SLI	Alert Severity	Action upon Breach
oldest_unacked_message_age	\$< 5\$ minutes	P1 (Critical)	Pages Data Platform SRE via ServiceNow webhook
dataflow/system_lag	\$< 120\$ seconds	P2 (Major)	Triggers automated worker auto-scaling
dataflow/watermark_age	\$< 180\$ seconds	P2 (Major)	Alerts on upstream source timestamp drift
dlq_message_count	\$< 100\$ msgs/hour	P3 (Warning)	Flags upstream schema violation for review

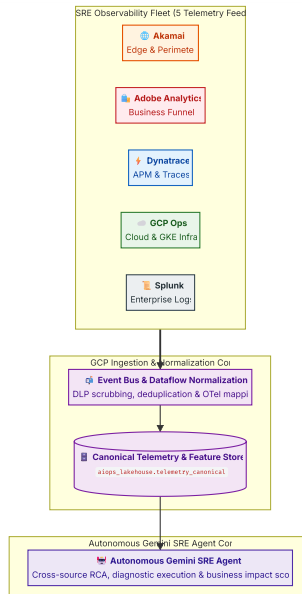
5. Related Architecture Specifications

- [SRE Observability Fleet - Source Telemetry Matrix](#)
- [Data Contracts & Canonical Schemas](#)
- [Ingestion Best Practices Guide](#)
- [Data Processing & AI Feature Store](#)

SRE Observability Fleet - Source Telemetry Matrix

1. Executive Overview

The **Autonomous Gemini SRE Agent** correlates telemetry across 5 core observability tools used by enterprise SRE teams. Instead of viewing these tools as isolated monitoring silos, the AIOps platform models them as high-dimensional feature feeds that map to specific operational and business domains:



2. Comprehensive Tool Telemetry Profiles

2.1 Akamai (Edge & Perimeter Observability)

- **Domain:** Edge CDN, DNS, Web Application Firewall (WAF), DDoS mitigation, and Bot Management.
- **Emitted Telemetry:**
 - Edge HTTP latency and Time to First Byte (TTFB).
 - HTTP status distributions (502/503/504 edge vs. origin error rates).
 - WAF security triggers, DDoS mitigation vectors, and Bot Manager classification scores.
- **Ingestion Channel:** Akamai **DataStream 2** HTTPS Push to Cloud Run Ingress → `telemetry.akamai.raw`.
- **Agent Reasoning Value:**
 1. Distinguishes regional ISP/CDN network degradation from core backend microservice crashes.
 2. Isolates bot-driven traffic spikes to prevent unnecessary container autoscaling.

2.2 Dynatrace (APM, Distributed Tracing & Topology)

- **Domain:** Full-stack Application Performance Monitoring (APM), code-level distributed tracing (PurePath), runtime dependency mapping (Smartscape), and Davis AI problem detection.
- **Emitted Telemetry:**
 - Distributed trace spans with OpenTelemetry headers (`trace_id` , `span_id`).
 - JVM/CLR thread pool saturation, Garbage Collection (GC) pauses, and memory heap metrics.
 - Smartscape entity topology DAGs and Davis AI root cause problem webhooks.
- **Ingestion Channel:** Real-time problem webhooks → `telemetry.dynatrace.raw` ; scheduled REST API sync for Smartscape DAGs (`/api/v2/entities`).
- **Agent Reasoning Value:**
 1. Directly identifies the failing class, method, or SQL database lock from PurePath stack traces.
 2. Traverses Smartscape dependency graphs to calculate upstream and downstream blast radius.

2.3 Google Cloud Operations Suite (Cloud & Platform Infrastructure)

- **Domain:** Native GCP infrastructure, Google Kubernetes Engine (GKE), serverless runtimes, and Cloud Audit Logs.
- **Emitted Telemetry:**
 - Node/Pod CPU and Memory saturation, container `CrashLoopBackOff` restart counts.
 - Cloud Pub/Sub unacknowledged message age and Dataflow system watermarks.
 - Cloud Audit Logs (identifying recent CI/CD deployments and configuration changes).
- **Ingestion Channel:** Zero-compute **Cloud Logging Log Router Sinks** → `telemetry.gcp.raw` ; Google Managed Service for Prometheus (GMP).
- **Agent Reasoning Value:**
 1. Correlates sudden application error spikes with recent Kubernetes deployments or configuration drift.
 2. Evaluates infrastructure resource starvation (OOM kills, CPU throttling).

2.4 Splunk (Enterprise Logging & SIEM Forensics)

- **Domain:** Enterprise centralized log repository, physical Point-of-Sale (POS) logs, middleware message queues (Kafka, IBM MQ), and SIEM events.
- **Emitted Telemetry:**
 - Unstructured application logs, stack traces, and database connection pool exceptions.
 - POS terminal hardware and peripheral transaction logs.
 - Splunk Enterprise Security (ES) notable security events.
- **Ingestion Channel:** Splunk **HTTP Event Collector (HEC)** → `telemetry.splunk.raw` ; on-demand REST API for time-windowed SPL queries (`$\pm 10\text{ minutes}$`).
- **Agent Reasoning Value:**
 1. Performs semantic error log clustering across dozens of microservices.
 2. Pulls forensic log context matching the incident's `trace_id` for inclusion in ServiceNow work notes.

2.5 Adobe Analytics (Digital Experience & Business Telemetry)

- **Domain:** Customer conversion funnels, e-commerce checkout telemetry, and digital revenue transactions.
- **Emitted Telemetry:**
 - Real-time Orders Per Minute (OPM), Cart Additions (`scAdd`), Checkout Funnel Drop-offs (`scCheckout`).
 - Payment gateway third-party failure responses.
 - User cohort attributes (geographic region, device type, app version).
- **Ingestion Channel:** Adobe Experience Platform (AEP) Streaming Ingestion → `telemetry.adobe.raw` .
- **Agent Reasoning Value:**
 1. **Silent Outage Detection:** Identifies frontend or payment bugs where technical infrastructure appears green but business transactions are failing.
 2. **Financial Impact Scoring:** Calculates real-time estimated revenue loss (USD/min) to drive accurate incident prioritization (P1 vs. P3) in ServiceNow.

3. Canonical Field Mapping Matrix (`telemetry_canonical`)

All 5 feeds are standardized into the OpenTelemetry-aligned canonical schema in BigQuery:

Canonical Field	Type	Akamai Mapping	Dynatrace Mapping	GCP Ops Mapping	Splunk Mapping	Adobe Mapping
<code>event_id</code>	STRING	<code>requestId</code>	<code>ProblemID</code> / <code>spanId</code>	<code>insertId</code>	<code>_cd</code> / <code>eventId</code>	<code>eventId</code>
<code>timestamp</code>	TIMESTAMP	<code>startEpoch</code>	<code>StartTime</code>	<code>timestamp</code>	<code>_time</code>	<code>time</code>
<code>source_tool</code>	STRING	<code>'akamai'</code>	<code>'dynatrace'</code>	<code>'gcp_ops'</code>	<code>'splunk'</code>	<code>'adobe'</code>
<code>severity</code>	STRING	Derived from HTTP status	<code>ProblemSeverity</code>	<code>severity</code>	Derived from log level	Derived from log level
<code>entity.service_name</code>	STRING	<code>property_name</code>	<code>ImpactedEntity</code>	<code>resource.labels.container_name</code>	<code>source</code> / <code>service</code>	<code>Funnel</code> / <code>check</code>
<code>entity.host</code>	STRING	<code>edge_ip</code>	<code>HostName</code>	<code>resource.labels.node_name</code>	<code>host</code>	N/A (Custom Browser)
<code>entity.cmdb_ci_id</code>	STRING	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping	CMDB CI mapping
<code>log_payload.trace_id</code>	STRING	<code>X-Trace-ID</code>	<code>PurePathTraceId</code>	<code>trace</code>	<code>trace_id</code>	<code>trace_id</code>
<code>log_payload.message</code>	STRING	Request URI & Error	Error message	<code>textPayload</code> / <code>jsonPayload</code>	<code>_raw</code>	N/A
<code>business_context.orders_per_minute</code>	FLOAT64	N/A	N/A	N/A	N/A	<code>orders_per_minute</code>
<code>business_context.estimated_revenue_loss_usd</code>	FLOAT64	N/A	N/A	N/A	N/A	Calculated

4. Cross-Domain Architectural Linkages

- [Ingestion Master Blueprint & Streaming Pipelines](#)
- [Data Contracts & Canonical Schemas](#)

- [Data Processing & AI Feature Store](#)
 - [Autonomous Gemini SRE Agent Architecture](#)
-

Telemetry Data Contracts, Canonical Schemas & DLP Specifications

1. Overview & Objectives

In a heterogeneous enterprise observability ecosystem, each tool emits telemetry in proprietary, non-standard formats (e.g., Dynatrace PurePath JSON, Akamai DataStream records, Splunk CIM, Adobe HIT clickstream, GCP LogEntry).

To enable centralized ML anomaly detection, Vertex AI LLM correlation, and cross-domain root-cause analysis, the Ingestion Layer enforces a **Canonical Telemetry Event Contract**. All incoming raw streams are mapped and validated into this unified schema inside **Cloud Dataflow** before writing to **BigQuery** or publishing to the **Vertex AI Reasoning Bus**.

2. Canonical Telemetry Event Schema

The Canonical Schema is formally defined in **Apache Avro** and **JSON Schema** formats, supporting backward and forward schema evolution.

2.1 JSON Schema Specification (`aiops.canonical.event.v1`)

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "title": "AIOpsCanonicalEvent",
  "type": "object",
  "required": ["event_id", "timestamp", "source_tool", "domain", "severity", "entity"],
  "properties": {
    "event_id": {
      "type": "string",
      "format": "uuid",
      "description": "Globally unique deterministic or UUIDv4 event identifier for deduplication."
    },
    "timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "UTC ISO-8601 timestamp when the event occurred at source."
    },
    "ingestion_timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "UTC ISO-8601 timestamp when GCP Dataflow processed the record."
    },
    "source_tool": {
      "type": "string",
      "enum": ["AKAMAI", "DYNATRACE", "GCP_OPS", "SPLUNK", "ADOBE_ANALYTICS"],
      "description": "The originating observability tool."
    },
    "domain": {
      "type": "string",
      "enum": ["EDGE_SECURITY", "APM_TRACES", "INFRASTRUCTURE", "ENTERPRISE_LOGS", "BUSINESS_TELEMETRY"],
      "description": "Functional observability domain."
    },
    "severity": {
      "type": "string",
      "enum": ["CRITICAL", "HIGH", "MEDIUM", "LOW", "INFORMATIONAL"],
      "description": "Normalized enterprise severity level."
    },
    "entity": {
      "type": "object",
      "required": ["service_name", "environment"],
      "properties": {
        "service_name": { "type": "string", "example": "checkout-api" },
        "environment": { "type": "string", "enum": ["production", "staging", "development"] },
        "host": { "type": "string", "example": "gke-prod-us-central1-pool-01" },
        "container_name": { "type": "string", "example": "checkout-service-pod-7d4f9" },
        "cloud_provider": { "type": "string", "enum": ["GCP", "AWS", "AZURE", "EDGE_AKAMAI", "ON_PREM"] },
        "region": { "type": "string", "example": "us-central1" },
        "cmdb_ci_id": { "type": "string", "description": "ServiceNow CMDB Configuration Item sys_id" }
      }
    },
    "metrics": {
      "type": "array",
      "items": {
        "type": "object",
        "required": ["name", "value", "unit"],
        "properties": {
          "name": { "type": "string", "example": "response_time_p99" },
          "value": { "type": "number", "example": 2450.5 },
          "unit": { "type": "string", "example": "ms" },
          "dimensions": { "type": "object", "additionalProperties": { "type": "string" } }
        }
      }
    },
    "log_payload": {
      "type": "object",

```

```

    "properties": {
      "message": { "type": "string", "description": "DLP-sanitized log message or exception string." },
      "stack_trace": { "type": "string", "description": "DLP-sanitized code call stack." },
      "trace_id": { "type": "string", "example": "4bf92f3577b34da6a3ce929d0e0e4736" },
      "span_id": { "type": "string", "example": "00f067aa0ba902b7" },
      "error_code": { "type": "string", "example": "HTTP_504_GATEWAY_TIMEOUT" }
    }
  },
  "business_context": {
    "type": "object",
    "properties": {
      "orders_per_minute": { "type": "number", "example": 420.0 },
      "cart_abandonment_rate": { "type": "number", "example": 0.185 },
      "estimated_revenue_loss_usd": { "type": "number", "example": 15400.0 },
      "funnel_stage": { "type": "string", "enum": ["BROWSE", "CART", "CHECKOUT", "PAYMENT"] },
      "user_cohort": { "type": "string", "example": "iOS_App_v18.2" }
    }
  },
  "raw_attributes": {
    "type": "object",
    "description": "Original raw tool attributes retained for forensics (DLP sanitized).",
    "additionalProperties": true
  }
}

```

3. Tool-to-Canonical Field Mapping Matrix

Canonical Field	Akamai Mapping	Dynatrace Mapping	GCP Ops Suite Mapping	Splunk Mapping	Adobe Analytics Mapping
<code>event_id</code>	<code>reqId</code>	<code>problemId</code> / PurePath Span ID	<code>insertId</code>	<code>_cd</code> / <code>event_id</code>	<code>hitid_high</code> + <code>hitid_low</code>
<code>timestamp</code>	<code>start</code> (epoch)	<code>startTime</code>	<code>timestamp</code>	<code>_time</code> (epoch)	<code>date_time</code> (UTC)
<code>source_tool</code>	"AKAMAI"	"DYNATRACE"	"GCP_OPS"	"SPLUNK"	"ADOBE_ANALYTICS"
<code>domain</code>	"EDGE_SECURITY"	"APM_TRACES"	"INFRASTRUCTURE"	"ENTERPRISE_LOGS"	"BUSINESS_TELEMETRY"
<code>severity</code>	Calculated from WAF action / HTTP 5xx	Davis AI Severity (<code>AVAILABILITY</code> , <code>PERFORMANCE</code>)	<code>severity</code> (<code>CRITICAL</code> , <code>ERROR</code>)	<code>urgency</code> / <code>alert_level</code>	Anomaly deviation magnitude (\$Z\$-score)
<code>entity.service_name</code>	<code>cpCode</code> / Host header	Dynatrace <code>entityName</code> / Service Tag	<code>resource.labels.container_name</code>	<code>source</code> / <code>index</code>	<code>reportSuiteID</code> / App ID
<code>metrics</code>	TTFB, Bytes Transferred, WAF rule score	Response Time (p95/p99), CPU %, GC pause ms	Container CPU, Memory limit %, Pub/Sub lag	Transaction volume, Search runtime	Orders Per Minute (OPM), Cart Adds, GMV
<code>log_payload.trace_id</code>	<code>edgeTraceId</code>	PurePath <code>traceId</code>	<code>trace</code> (<code>projects/ ... /traces/ ...</code>)	CIM field <code>trace_id</code>	<code>marketing_cloud_visitor_id</code>
<code>business_context</code>	Geolocation, ISP, Bot Score	User Session tags, Cart Value	N/A	POS store ID, Terminal ID	Orders/Min, Cart Drop %, Funnel stage

4. Cloud DLP (Data Loss Prevention) Sanitization Rules

To ensure strict compliance with **PCI-DSS**, **GDPR**, and **CCPA**, all incoming text payloads, log messages, stack traces, and query strings pass through **Google Cloud Data Loss Prevention (Cloud DLP)** inside the Dataflow pipeline before persistent storage in BigQuery.

4.1 In-Flight DLP De-Identification Templates



4.2 Configured InfoTypes & Transformation Strategies

InfoType Category	Detected Patterns	Transformation Method	Output Format in BigQuery
Credit Card Numbers	Visa, MasterCard, Amex, Discover	Masking: Mask all characters except last 4	*****1234
CVV / CVV2	3-4 digit card security codes	Complete Redaction	[CVV_REDACTED]
Email Addresses	User email headers and query strings	Crypto-Deterministic Hash (HMAC-SHA256)	hash_e4a8b71 ... (enables cross-session cohorting without PII exposure)
Phone Numbers	US/International phone formats	Character Replacement	[PHONE_REDACTED]
Authentication Tokens	Bearer tokens, API Keys, JWT signatures	Regex Redaction	Bearer [TOKEN_REDACTED]
Passwords / Passphrases	Form body passwords, connection strings	String Replacement	password=[REDACTED]

5. Schema Evolution & Governance

- Forward Compatibility:** New optional fields can be added without breaking existing Dataflow stream parsers.
- Schema Registry:** Canonical Avro schemas are stored and versioned in a centralized GCP Cloud Storage schema registry bucket (`gs://aiops-schema-registry/v1/`).
- Validation Enforcement:** Messages failing canonical JSON Schema validation are rejected by Dataflow and directed to the Dead-Letter Queue (`telemetry.<source>.dlq`) with an attached `validation_error_code` .

AIOps Ingestion Pipeline: Best Practices

This guide outlines the best practices for designing, configuring, and operating the AIOps ingestion pipelines based on the core architecture defined in [ingestion_architecture.md](#). Adhering to these guidelines ensures scalability, security, and high reliability across multi-cloud observability sources.

1. Design Best Practices

1.1 Connector Pattern Selection

Always select the ingestion connector pattern based on the source's capabilities and network location:

- **Use Native Connectors (Cloud Logging/Log Router Sinks)** for any workloads already running on Google Cloud (e.g., GKE, Dataflow). This guarantees sub-second latency and zero egress costs.
- **Use Push Connectors (Cloud Run)** for modern SaaS platforms (Akamai, Dynatrace, Adobe, Splunk HEC) that support real-time webhooks or streaming exports.
- **Use Pull Connectors (Cloud Composer / Apache Airflow)** *only* for legacy on-premise databases or air-gapped systems that cannot push data out. Avoid using Cloud Run jobs for Pull scenarios, as Airflow provides superior retry logic and operational visibility for complex extraction tasks.

1.2 Strict Decoupling

- **Never connect sources directly to processing engines.** All incoming telemetry must land in **Cloud Pub/Sub** first.
 - Use isolated, dedicated Pub/Sub topics for each major data source (e.g., `telemetry.akamai.raw`, `telemetry.splunk.raw`). This prevents a traffic flood from one source (like a DDoS attack logged by Akamai) from starving resources for other critical telemetry.
-

2. Configuration Best Practices

2.1 Security & Authentication

- **Never hardcode credentials.** Cloud Run push connectors and Cloud Composer pull tasks must retrieve API keys, HMAC secrets, and database credentials dynamically from **GCP Secret Manager** at runtime.
- Protect all public-facing Cloud Run push connectors with **Cloud Armor WAF**. Enforce strict IP allowlisting (e.g., only accepting traffic from known Akamai or Dynatrace CIDR blocks) and implement rate-limiting to prevent malicious flooding.

2.2 Schema Management

- Enable and enforce the **Pub/Sub Schema Registry** using Protobuf or Avro.
- Connectors must validate JSON payloads against the registry before publishing. This prevents upstream SaaS providers from pushing undocumented, breaking schema changes that crash downstream Dataflow pipelines.

2.3 Resiliency Configuration

- **Multi-Region Pub/Sub:** Configure critical telemetry topics for cross-region message routing so that data ingestion survives single-zone or single-region outages.
 - **Dataflow Auto-scaling:** Configure Dataflow streaming jobs with a generous `maxNumWorkers` limit. This allows the pipeline to gracefully absorb sudden spikes in error logs during a massive P1 outage.
-

3. Operational & Observability Best Practices

3.1 Dead-Letter Queues (DLQs)

- **Never silently drop data.**
- Configure Pub/Sub native DLQs (after 5 failed delivery attempts) to catch messages that repeatedly crash Dataflow workers.
- Configure Dataflow side-outputs to catch messages that parse successfully but fail business logic validation (e.g., missing timestamps).
- Route all DLQ streams to a dedicated **Cloud Storage (GCS) Bucket** partitioned by date (`gs://aiops-dlq-bucket/YYYY/MM/DD/`) so data engineers can inspect and replay the data later.

3.2 Key Service Level Indicators (SLIs) to Monitor

You must configure **Cloud Monitoring** dashboards and alerts for the following critical ingestion SLIs:

- **Pub/Sub `oldest_unacked_message_age`** : If this exceeds 5 minutes, it indicates the Dataflow pipeline is stalled or falling behind.
- **Dataflow `system_lag` and `data_watermark_age`** : Monitors the real-time processing delay.
- **DLQ Bucket Depth:** A sudden spike in DLQ volume indicates an upstream schema change or a bad code deployment in the Dataflow parsing logic.

3.3 Incident Alerting Routing

- Route all critical ingestion pipeline alerts directly to **ServiceNow** via Cloud Monitoring webhooks.

- If the AIOps platform itself is failing to ingest data, the internal Data Platform SRE team must be paged immediately, as "silent monitoring failures" are the highest risk to enterprise uptime.
-

Unified Lakehouse & Analytical Storage Architecture

1. Overview & Objectives

The **AIOps Data Lakehouse** provides a unified, highly optimized analytical storage layer built natively on **Google Cloud Platform (GCP)**. It serves three distinct workloads essential to the AIOps Intelligence Layer:

1. **Low-Latency Analytical Querying:** Supporting real-time ML feature extraction, BQML anomaly detection models, and Gemini SRE Agent contextual queries.
2. **Cost-Effective Long-Term Telemetry Archival:** Storing petabyte-scale historical logs and traces in compressed **Apache Parquet** format for historical trend analysis and compliance auditing.
3. **Semantic Retrieval-Augmented Generation (RAG):** Storing vectorized representations of Runbooks (SOPs) and historical incident data for real-time retrieval by the Gemini SRE Agent.



Syntax error in text
mermaid version 10.9.8

2. BigQuery Dataset & Table Design

2.1 Primary Analytical Table: `telemetry_canonical`

- **Table Type:** Partitioned & Clustered Columnar Table.
- **Partitioning Specification:** Partition by day on `DATE(timestamp)`.
 - Partition Expiration: Optional (managed via lifecycle tiering).
- **Clustering Keys:** `source_tool`, `entity.service_name`, `severity`, `domain`.
- **Rationale:** SRE forensic queries and ML anomaly jobs almost always filter by time range, tool, and service name. Clustering reduces bytes scanned by over 85%, ensuring fast query response times and minimal BigQuery query costs.

2.2 Table Schema DDL (BigQuery SQL)

```
CREATE OR REPLACE TABLE `aiops_lakehouse.telemetry_canonical`
(
  event_id STRING NOT NULL,
  timestamp TIMESTAMP NOT NULL,
  ingestion_timestamp TIMESTAMP NOT NULL,
  source_tool STRING NOT NULL,
  domain STRING NOT NULL,
  severity STRING NOT NULL,
  entity STRUCT<
    service_name STRING,
    environment STRING,
    host STRING,
    container_name STRING,
    cloud_provider STRING,
    region STRING,
    cmdb_ci_id STRING
  >,
  metrics ARRAY<STRUCT<
    name STRING,
    value FLOAT64,
    unit STRING,
    dimensions JSON
  >>,
  log_payload STRUCT<
    message STRING,
    stack_trace STRING,
    trace_id STRING,
    span_id STRING,
    error_code STRING
  >,
  business_context STRUCT<
    orders_per_minute FLOAT64,
    cart_abandonment_rate FLOAT64,
    estimated_revenue_loss_usd FLOAT64,
    funnel_stage STRING,
    user_cohort STRING
  >,
  raw_attributes JSON
)
PARTITION BY DATE(timestamp)
CLUSTER BY source_tool, entity.service_name, severity;
```

2.3 Real-Time AI Feature Store & Materialized Views

To power the **ARIMA_PLUS** models in **BigQuery ML (BQML)** without scanning the massive **telemetry_canonical** table every minute, we utilize **BigQuery Materialized Views**. These act as a continuously updated Feature Store for the ML engine.

For deep-dive specifications on time-series resampling, sliding window alert clustering, trace sessionization, and topology graph construction, see the dedicated [Data Processing & AI Feature Preparation Architecture](#).

By pre-aggregating high-frequency metrics (like Orders Per Minute), the Semantic Router and BQML can evaluate anomalies in sub-seconds with near-zero cost.

```
-- Materialized View for 1-minute OPM (Orders Per Minute) Rollups
CREATE MATERIALIZED VIEW `aiops_lakehouse.telemetry_opm_1min_mv`
PARTITION BY DATE(window_start)
CLUSTER BY service_name
AS
SELECT
  TIMESTAMP_TRUNC(timestamp, MINUTE) AS window_start,
  entity.service_name,
  SUM(business_context.orders_per_minute) AS total_opm,
  AVG(business_context.cart_abandonment_rate) AS avg_abandon_rate
FROM
  `aiops_lakehouse.telemetry_canonical`
WHERE
  source_tool = 'adobe_analytics'
  AND business_context.orders_per_minute IS NOT NULL
GROUP BY
  window_start,
  service_name;
```

3. Storage Tiering & Lifecycle Management

Storage Tier	Data Age	Storage Technology	Storage Cost Profile	Primary Query Engine
Hot Tier	0 – 30 Days	BigQuery Active Storage	Standard BigQuery Active Storage	BigQuery SQL / Vertex AI Agent
Warm Tier	31 – 365 Days	BigLake (Iceberg / BigQuery Long-Term)	50% discount on BigQuery storage pricing	BigQuery SQL / Dataproc Serverless
Cold Tier	1 – 7 Years	Google Cloud Storage (Archive Class)	Ultra-low cost object storage (\$0.0012/GB/mo)	BigQuery BigLake External Queries

4. Vector Lakehouse & RAG Storage

To support the **Context-Aware Incident Response** requirement of the Intelligence Layer, unstructured text must be semantically searchable. This requires a dedicated Vector Lakehouse architecture parallel to the relational BigQuery tables.

4.1 Storage Pipeline

- Raw Storage:** Markdown Runbooks (SOPs) are managed in GitHub ("Runbooks as Code"). A GitHub Actions pipeline syncs merged Markdown files directly to a standard **Google Cloud Storage (GCS) Bucket**.
- Embedding:** A Cloud Function triggers on new GCS files, chunks the Markdown, and calls the **Vertex AI Text Embeddings API** to generate high-dimensional vectors.
- Vector Database:** The resulting vectors, along with metadata (e.g., `service_name: checkout-service` , `author: sre-pod-1`), are indexed in **Vertex AI Vector Search**.

4.2 Retrieval

When the Gemini SRE Agent begins diagnosing a P1 incident, it extracts the `service_name` and `symptoms` from the alert payload. It executes an Approximate Nearest Neighbor (ANN) search against Vertex AI Vector Search to retrieve the exact Markdown SOP required for remediation within milliseconds.

5. Integration with the Autonomous Gemini SRE Agent

The **Autonomous Gemini SRE Agent** relies heavily on the Hot Tier and Materialized Views of this Lakehouse.

When an incident signature is triggered, the Agent uses **Function Calling (Tools)** to query BigQuery securely:

- Tool_Topology_Graph** : Traverses the `topology_service_graph` table to inspect upstream caller and downstream dependency trees.
- Tool_Diagnostic_Sandbox** : Executes AST-safe SQL queries against `aiops_lakehouse.telemetry_canonical` to correlate concurrent Splunk error logs and Dynatrace PurePaths matching the incident's `trace_id` (`$\pm 10\text{ minutes}`).
- Materialized Baseline Queries:** Reads `telemetry_opm_1min_mv` via BigQuery BI Engine in `< 500\text{ms}` to verify if technical alerts correspond to active digital revenue drop-offs.

6. High Availability & Disaster Recovery (Resiliency)

To ensure the AIOps platform remains available during regional outages, the storage layer is designed for maximum resilience:

1. **Multi-Region Deployments:** The BigQuery `aiops_lakehouse` dataset and all GCS Cold Tier buckets are configured as **Multi-Region** (e.g., `US` or `EU`). This ensures that if a single GCP region (like `us-central1`) experiences an outage, the SRE team retains access to all historical telemetry and ML models to diagnose the outage.
 2. **Vertex AI High Availability:** The Vector Search index is deployed with multiple replicas across distinct zones within the region to guarantee high QPS availability for the Gemini SRE Agent.
-

7. Security, Governance & PII Compliance

Retail telemetry often inadvertently captures sensitive data. The Lakehouse implements strict defense-in-depth:

1. **Cloud DLP Masking:** Before writing to BigQuery, the Dataflow ingestion pipeline utilizes the Cloud Data Loss Prevention (DLP) API to automatically detect and redact PII (e.g., credit card numbers, email addresses) from log payloads.
 2. **Row-Level Security (RLS):** BigQuery Row-Level Security policies are applied to `telemetry_canonical` so that SRE pods can only query telemetry for the specific microservices (e.g., `service_name = 'checkout-service'`) they own.
 3. **Encryption at Rest (CMEK):** All data stored in BigQuery, Cloud Storage, and Vertex AI is encrypted at rest using **Customer-Managed Encryption Keys (CMEK)** via Cloud KMS, fulfilling strict retail compliance mandates.
-

8. Performance & Cost Optimization Best Practices

1. **Parquet Columnar Format:** All batched data files written to Cloud Storage utilize **Apache Parquet with Snappy or ZSTD compression**, optimizing storage footprint and scan efficiency.
 2. **Partition Pruning:** All analytical queries generated by the Gemini SRE Agent and Semantic Router must include mandatory time range filters (`WHERE timestamp ≥ TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 1 HOUR)`).
 3. **BigQuery BI Engine:** Enabled with a 50 GB memory reservation to accelerate real-time Materialized View reads and high-frequency anomaly evaluation queries by BQML.
-

9. Related Architectural Specifications

- [Data Processing & AI Feature Preparation](#)
 - [AIOps Intelligence Layer Specification](#)
 - [Ingestion Master Blueprint](#)
-

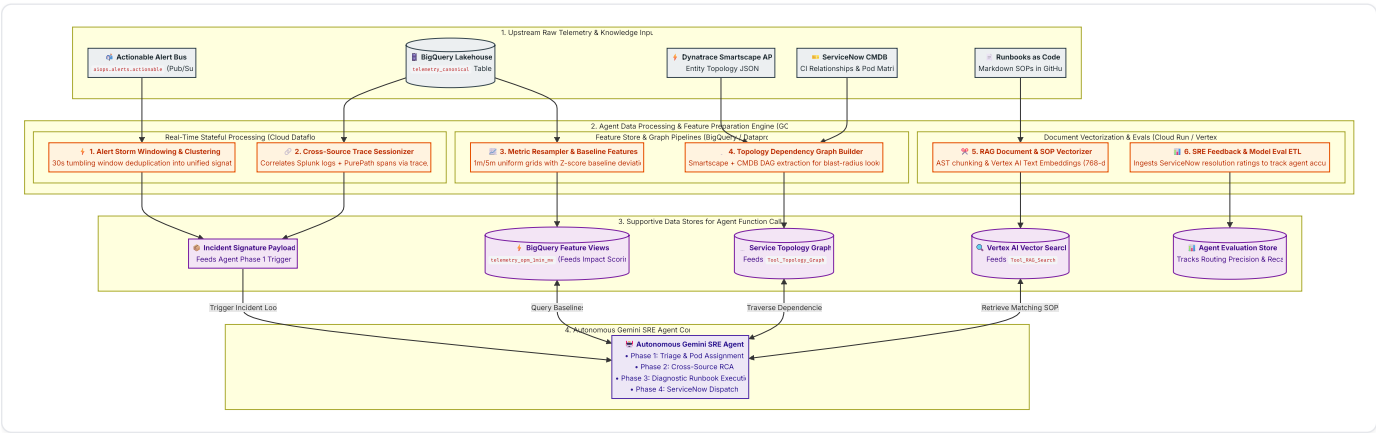
Data Processing & AI Feature Preparation Architecture

1. Executive Summary & Objective

The **Data Processing & AI Feature Preparation Layer** serves as the critical data engineering backbone for the **Autonomous Gemini SRE Agent**, hosted natively on **Google Cloud Platform (GCP)**.

Raw telemetry streams into the platform at rates between **150,000 and 500,000+ Events Per Second (EPS)**. Large Language Models (LLMs) cannot consume raw, high-cardinality event streams directly due to context window limits, token latency, and noise.

This layer implements stateful stream processing, topological graph extraction, time-series resampling, document vectorization, and evaluation ETL to synthesize raw telemetry into high-value **supportive data structures** directly consumed by the agent's function-calling tools:



2. The 6 Agent-Supportive Data Pipelines

2.1 Pipeline 1: Stateful Alert Storm Windowing & Incident Clustering

- Agent Support Role:** Generates the structured **Incident Signature JSON** that triggers **Phase 1 (Triage & Assignment)** of the Gemini SRE Agent.
- Mechanism (Cloud Dataflow / Apache Beam):**
 - Buffers high-priority incoming alerts (**severity** $\geq$ **WARN**) in a **30-second tumbling window**.
 - Groups cascading alerts across tools (Akamai 504s, GKE pod restarts, Dynatrace problems, Splunk database errors) by shared topological keys (**entity.service_name** or **entity.cmdb_ci_id**).
 - Synthesizes 50+ raw alert events into a compact, token-efficient JSON payload:

```
{
  "incident_signature_id": "sig-2026-08-26-chk-001",
  "window_start": "2026-08-26T06:30:00Z",
  "window_end": "2026-08-26T06:30:30Z",
  "primary_impacted_service": "checkout-service",
  "alert_count": 48,
  "sources_involved": ["akamai", "dynatrace", "splunk", "gcp_ops"],
  "symptoms": [
    "Akamai: HTTP 504 rate = 14.2% on /api/v2/checkout",
    "Dynatrace: Thread pool starvation in CheckoutService.processPayment()",
    "Splunk: 320 DB connection timeout exceptions on spanner-prod-01",
    "GCP Ops: Pod memory pressure warning on gke-checkout-pod-x89"
  ],
  "correlated_trace_ids": ["9f8a7b6c5d4e3f2a1b0c9d8e7f6a5b4c"]
}
```

2.2 Pipeline 2: Metric Resampling & Fast Baseline Feature Store

- Agent Support Role:** Provides the agent with sub-second statistical baseline lookups during **Phase 2 (Deep RCA & Financial Impact Scoring)**.
- Mechanism (BigQuery Materialized Views & BI Engine):**
 - Aggregates high-frequency business clickstream and infrastructure telemetry into strict **1-minute** and **5-minute** time buckets.
 - Calculates rolling 7-day seasonal averages and **\$z\$-score** deviation baselines.
 - Enabled with BigQuery BI Engine to allow the Gemini Agent to evaluate whether a technical alert corresponds to real business revenue loss in **\$ < 500**.

```
-- Production Feature View for Real-Time Anomaly Baseline Detection
CREATE MATERIALIZED VIEW `aiops_lakehouse.telemetry_opm_1min_mv`
PARTITION BY DATE(window_start)
CLUSTER BY service_name
AS
SELECT
  TIMESTAMP_TRUNC(timestamp, MINUTE) AS window_start,
  entity.service_name,
  SUM(COALESCE(business_context.orders_per_minute, 0.0)) AS total_opm,
  AVG(COALESCE(business_context.cart_abandonment_rate, 0.0)) AS avg_abandon_rate,
  COUNT(1) AS raw_event_count
FROM
  `aiops_lakehouse.telemetry_canonical`
WHERE
  source_tool = 'adobe_analytics'
GROUP BY
  window_start,
  service_name;
```

2.3 Pipeline 3: Dynamic Topology Graph ETL & CMDB Traversal

- **Agent Support Role:** Directly powers `Tool_Topology_Graph`, enabling the agent to identify responsible SRE pods and upstream/downstream blast radius.
- **Mechanism:**
 1. A scheduled Cloud Composer DAG extracts dependency trees hourly from **Dynatrace Smartscape API** (`/api/v2/entities`) and **ServiceNow CMDB** (`cmdb_rel_ci`).
 2. Persists relationships into an adjacency graph table in BigQuery:

```
CREATE TABLE `aiops_lakehouse.topology_service_graph` (
  parent_ci STRING NOT NULL,
  parent_service_name STRING NOT NULL,
  child_ci STRING NOT NULL,
  child_service_name STRING NOT NULL,
  relation_type STRING NOT NULL, -- 'CALLS', 'RUNS_ON', 'DEPENDS_ON'
  sre_owner_pod STRING NOT NULL,
  last_synced_at TIMESTAMP NOT NULL
)
PARTITION BY DATE(last_synced_at)
CLUSTER BY parent_service_name, child_service_name;
```

2.4 Pipeline 4: Cross-Source Trace & Log Sessionizer

- **Agent Support Role:** Prepares correlated forensic evidence blocks consumed by `Tool_Diagnostic_Sandbox`.
- **Mechanism:**
 1. OpenTelemetry headers (`trace_id`, `span_id`) propagated through HTTP requests are indexed upon ingestion.
 2. When an incident is triggered, the pipeline queries BigQuery across both Splunk error logs and Dynatrace PurePaths within a `$\pm 10\text{minute}` window matching `trace_id`.
 3. Formats the correlated trace stack and database query text as markdown evidence for the agent.

2.5 Pipeline 5: RAG Document Vectorization & Runbook ETL

- **Agent Support Role:** Builds and synchronizes the vector embeddings powering `Tool_RAG_Search`.
- **Mechanism:**
 1. Markdown SOPs managed in Git ("Runbooks as Code") are parsed via an AST parser upon GitHub push.
 2. Frontmatter metadata, symptom descriptions, and diagnostic commands are separated into structured chunks (max 512 tokens with overlap).
 3. Vectors are generated using the **Vertex AI Text Embeddings API** (`text-embedding-005`, 768 dimensions) and indexed in **Vertex AI Vector Search** for sub-100ms Approximate Nearest Neighbor (ANN) retrieval.

```
---
sop_id: SOP-CHK-DB-001
service_name: checkout-service
cldb_ci: CI-CHK-SVC-01
owner_pod: Checkout-Backend-Pod
tags: [spanner, connection_pool, timeout, 504]
diagnostics:
  - name: check_spanner_sessions
    type: bigquery_sql
    query: "SELECT host, count(1) as active_conn FROM `aiops_lakehouse.telemetry_canonical` WHERE source_tool = 'splunk' AND log_payload.message LIKE
---

# Spanner Connection Pool Exhaustion Runbook

## Symptoms
* Sudden spike in HTTP 504 errors from Akamai.
* Dynatrace flags Spanner client connection wait times > 5000ms.

## Remediation Steps
1. Verify active database session locks via Spanner admin console.
2. Scale up checkout service container replicas by 20%.
```

2.6 Pipeline 6: Closed-Loop SRE Feedback & Evaluation Dataset Preparation

- **Agent Support Role:** Captures human ground-truth feedback from ServiceNow to power continuous prompt tuning and agent evaluation.
- **Mechanism:**
 1. When an SRE closes or resolves an incident, ServiceNow triggers a webhook with engineer accuracy ratings (1–5 stars), confirmed root causes, and false-positive flags.
 2. An automated BigQuery SQL job joins the incident's original AI diagnosis with the human post-mortem resolution, streaming records into `aiops_lakehouse.agent_evaluations`.
 3. Evaluates rolling 7-day Routing Precision (\$> 92%\$) and SOP Retrieval Recall@3 (\$> 88%\$).

3. Data Processing SLA, Resiliency & Tool Mapping

Data Pipeline	Target Latency (SLA)	Compute Engine	Supported Agent Capability / Tool
1. Alert Storm Windowing	\$< 35\$ seconds	Cloud Dataflow (Streaming)	Agent Incident Trigger (<code>Incident Signature JSON</code>)
2. Metric Resampling	Sub-second read	BigQuery Materialized Views + BI Engine	Financial Impact Scoring (Phase 2 RCA)
3. Topology Graph Sync	\$< 10\$ minutes	Cloud Composer (Airflow) + BigQuery	<code>Tool_Topology_Graph</code> (Pod Assignment & Blast Radius)
4. Trace Sessionizer	\$< 2\$ seconds	BigQuery Clustered Query	<code>Tool_Diagnostic_Sandbox</code> (Cross-source correlation)
5. SOP Vectorization	\$< 60\$ seconds on Git Push	Cloud Functions + Vertex Embeddings	<code>Tool_RAG_Search</code> (Runbook & Post-Mortem Lookup)
6. Feedback Loop ETL	Hourly batch	BigQuery Scheduled Queries	Agent Evaluation & Continuous Learning

4. Summary & Cross-Domain Linkages

This data processing specification directly interfaces with the following modules:

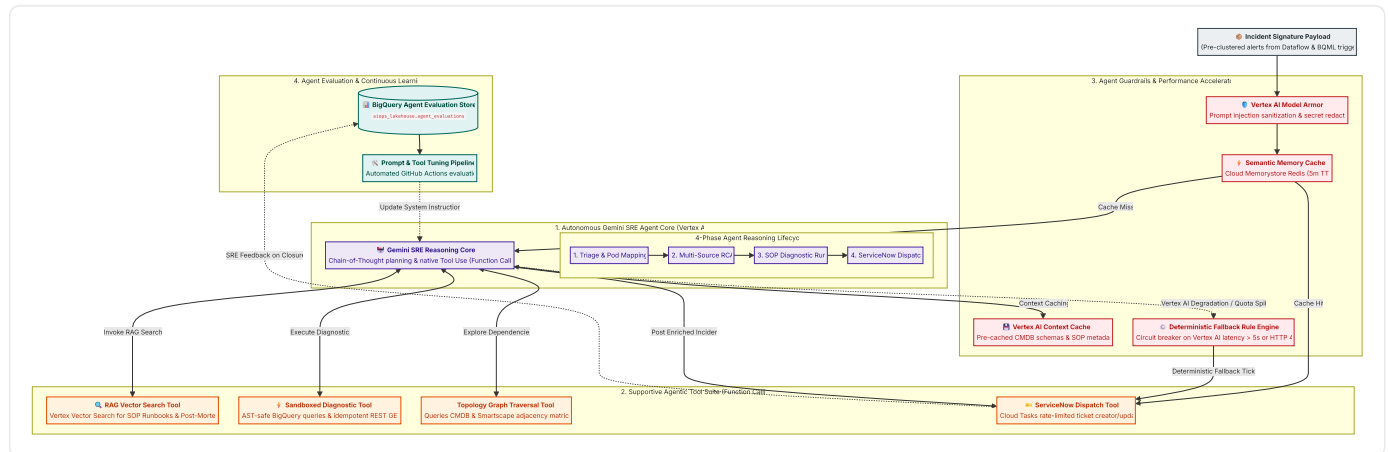
- [High-Level Platform Architecture](#)
- [SRE Observability Fleet - Source Telemetry Matrix](#)
- [Autonomous Gemini SRE Agent Architecture](#)
- [ServiceNow Integration & Remediation Guide](#)

AIOps Intelligence & Reasoning Layer - Agent-Centric Architecture

1. Executive Summary & Architectural Role

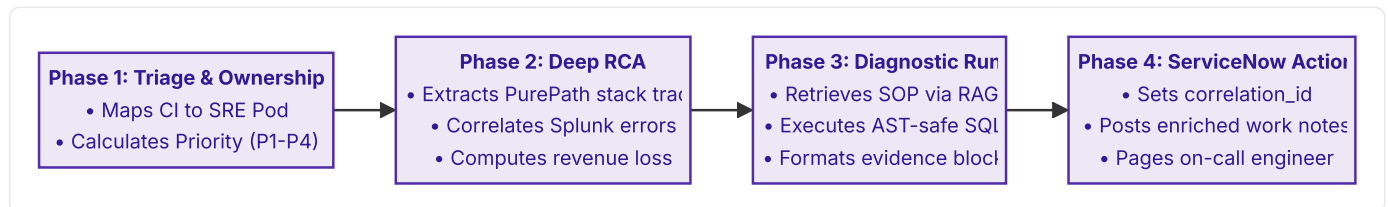
The **AIOps Intelligence & Reasoning Layer** is the cognitive core of the platform, powered natively by **Google Cloud Platform (Vertex AI & Gemini Models)**. Instead of deploying fragmented LLMs and disconnected execution engines, this layer centers around a unified **Autonomous Gemini SRE Agent** equipped with a specialized **Supportive Agentic Tool Suite**, **Semantic Memory**, **Model Armor Guardrails**, and **Context Caching**.

The agent consumes structured **Incident Signature Payloads** emitted by the [Data Processing & Feature Preparation Layer](#) and autonomously executes a 4-phase reasoning plan to triage incidents, isolate root causes, run non-destructive diagnostics, and update **ServiceNow**.



2. Autonomous Gemini SRE Agent: The 4-Phase Reasoning Lifecycle

Rather than using separate, uncoordinated LLMs for triage and investigation, the **Gemini SRE Agent** executes a single, cohesive reasoning loop via function calling:



Phase 1: Incident Triage & Pod Assignment

- **Objective:** Ingests the structured **Incident Signature Payload** and evaluates impact.
- **Tool Invocation:** Calls **Tool_Topology_Graph** to inspect parent-child microservice dependencies.
- **Outcome:** Assigns urgency, severity, and the responsible SRE pod (e.g., **Checkout-Backend-Pod** vs. **Network-Edge-Pod**), completely bypassing manual L1 helpdesk queues.

Phase 2: Multi-Source Root Cause Analysis (RCA)

- **Objective:** Correlates technical telemetry with business performance.
- **Actions:**
 1. Identifies the failing code path or database session via Dynatrace PurePath trace IDs.
 2. Synthesizes concurrent Splunk server error logs matching the incident time window (± 10 minutes).
 3. Estimates real-time digital financial impact (e.g., *"\$14.2k/min revenue drop based on Adobe Orders Per Minute deviation"*).

Phase 3: RAG Retrieval & Sandboxed Diagnostic Execution

- **Objective:** Autonomous execution of standard diagnostic runbooks before engineer engagement.
- **Actions:**
 1. Calls **Tool_RAG_Search** to retrieve the relevant Standard Operating Procedure (SOP) from **Vertex AI Vector Search**.
 2. Extracts all read-only diagnostic checks from the SOP YAML frontmatter.
 3. Invokes **Tool_Diagnostic_Sandbox** to execute the queries safely against BigQuery and internal monitoring APIs.

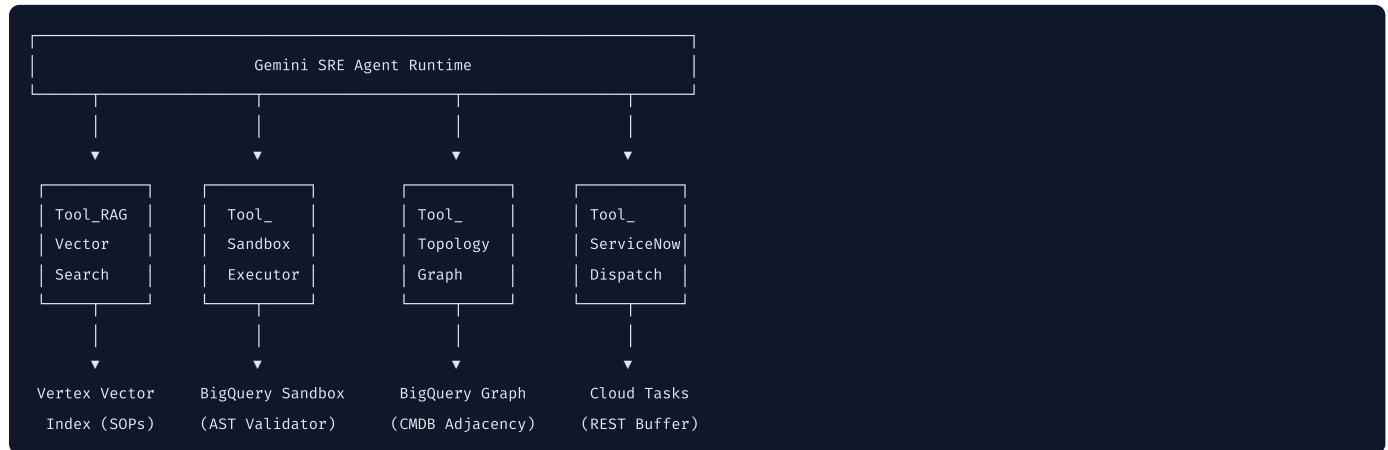
Phase 4: ServiceNow Dispatch & Action

- **Objective:** Publish the complete synthesized diagnosis to the enterprise system of record.

- **Actions:**
 1. Constructs the Markdown summary and diagnostic findings block.
 2. Invokes `Tool_ServiceNow_Dispatch` to create or update the incident in ServiceNow with `correlation_id = signature_id`.
 3. Automatically alerts the on-call engineer with complete diagnostic evidence pre-attached.

3. The Supportive Agentic Tool Suite (Function Calling)

The Gemini SRE Agent interacts with the enterprise ecosystem through four strictly governed tools:



3.1 `Tool_RAG_Search` (Runbook & Post-Mortem Retrieval)

- **Underlying Service:** Vertex AI Vector Search Index.
- **Capabilities:** Executes Approximate Nearest Neighbor (ANN) vector search (768-dimension embeddings) filtered by `service_name` and `environment`.
- **Output:** Returns the top-matching Markdown SOP containing symptoms, diagnostic commands, and remediation runbook links.

3.2 `Tool_Diagnostic_Sandbox` (Read-Only Execution Environment)

- **Underlying Service:** BigQuery Client + Cloud Run Gateway with AST Validator.
- **Guardrails:**
 - **AST SQL Parsing:** Blocks mutating statements (`INSERT` , `UPDATE` , `DELETE` , `DROP` , `ALTER` , `TRUNCATE` , `MERGE`). Only `SELECT` queries with mandatory `LIMIT ≤ 100` are permitted.
 - **HTTP Gate:** Restricts external diagnostic API calls strictly to idempotent `GET` requests.
 - **Quotas:** Hard 15-second execution timeout and maximum 1 GB BigQuery scan limit.

3.3 `Tool_Topology_Graph` (Dependency Traversal)

- **Underlying Service:** BigQuery `aiops_lakehouse.topology_service_graph`.
- **Capabilities:** Executes recursive BFS traversal across Dynatrace Smartscape and ServiceNow CMDB CI relationships to return upstream caller and downstream dependency trees.

3.4 `Tool_ServiceNow_Dispatch` (ITSM Enriched Ticket Publisher)

- **Underlying Service:** Asynchronous Cloud Tasks Queue (`aiops-snow-dispatch-queue`).
- **Capabilities:** Submits enriched JSON payloads to ServiceNow with rate-limiting protection (`$20\text{ req/s}`) and `correlation_id` deduplication.

4. Agent Guardrails, Security & Performance Acceleration

4.1 Vertex AI Model Armor & Input Sanitization

All raw telemetry strings and external API responses pass through an input sanitizer before entering the Gemini context window:

1. **Prompt Injection Defense:** Strips delimiter attacks (e.g., `Ignore previous instructions` , `SYSTEM PROMPT:` , `Assistant:`).
2. **Secret Redaction:** Regex tokenizers scrub API tokens, private keys, and passwords.
3. **Cloud DLP Integration:** Masks credit card numbers (PCI-DSS) and customer PII from log excerpts.

4.2 Vertex AI Context Caching

- **Mechanism:** Static CMDB microservice schemas, SRE pod matrices, system instructions, and tool definitions are pre-cached in Vertex AI Context Storage.
- **Impact:**
 - **Latency:** Reduces prompt token processing time by up to **80%**.
 - **Cost:** Decreases per-incident input token billing by up to **75%**.

4.3 Semantic Memory Cache (Cloud Memorystore Redis)

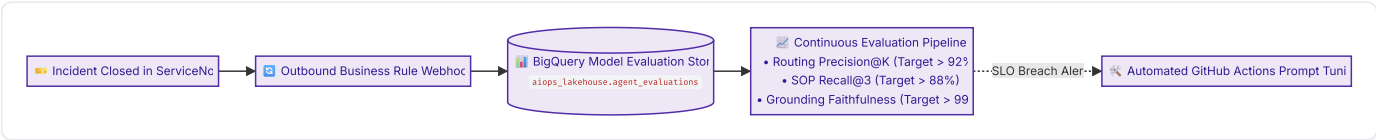
- **Mechanism:** Generates an embedding hash of the Incident Signature and queries a 5-minute TTL Redis cache.
- **Benefit:** Reuses RCA and diagnostic findings for recurring or duplicate alert bursts, avoiding redundant LLM invocations during major incidents.

4.4 Deterministic Rule Fallback Engine (Circuit Breaker)

- **Trigger Condition:** If Vertex AI experiences regional degradation, API latency spikes ($> 5\text{ seconds}$), or HTTP 429 quota exhaustion.
- **Action:** The circuit breaker trips immediately. Telemetry is routed through a deterministic Python rule engine using cached CMDB topology tables to assign the ticket, guaranteeing 100% incident routing uptime.

5. Closed-Loop SRE Feedback & Model Observability

To prevent model hallucination drift and continuously improve diagnostic quality, the agent operates in a continuous learning loop:



5.1 Evaluation Metrics & Quality Targets

- **Routing Accuracy ($> 92\%$):** Percentage of incidents assigned to the correct SRE pod without human re-assignment.
- **SOP Retrieval Recall@3 ($> 88\%$):** Percentage of incidents where the relevant remediation runbook was among the top-3 retrieved.
- **Grounding & Faithfulness ($> 99\%$):** Verification that all facts, metrics, and trace IDs in the ticket summary are grounded in raw telemetry.

6. Summary & Cross-Domain Linkages

This agent-centric specification interfaces directly with:

- [Platform Overview & End-to-End Blueprint](#)
- [Data Processing & Feature Preparation Architecture](#)
- [Unified Lakehouse Architecture](#)
- [ServiceNow Integration & Remediation Guide](#)

ServiceNow ITSM Integration & Automated Remediation Architecture

1. Overview & Architectural Role

ServiceNow serves as the definitive enterprise system of record for IT Service Management (ITSM), Configuration Management Database (CMDB), on-call scheduling, and automated incident response workflows.

The **ServiceNow Integration Layer** bridges the **Google Cloud Platform (GCP) AIOps Core** with enterprise operational workflows. It performs:

1. **Intelligent Direct Pod Routing:** Bypasses manual L1 triage by routing enriched incidents directly to specialized SRE pods based on CMDB topology and ML classification.
2. **Contextual Incident Enrichment:** Automatically attaches synthesized Dynatrace root cause traces, Splunk log forensics, and Adobe revenue loss calculations to work notes.
3. **Automated SOP Diagnostics Attachment:** Pre-executes read-only diagnostic checks and posts structured evidence to tickets before an engineer begins investigation.
4. **Stateful Incident Deduplication:** Correlates ongoing alert storms using unique `correlation_id` signatures to update active incidents rather than spamming duplicate tickets.
5. **Human-in-the-Loop Remediation:** Orchestrates one-click remediation actions via ServiceNow Flow Designer calling back to sandboxed GCP remediation runners.
6. **Closed-Loop SRE Feedback ETL:** Captures post-incident resolution ratings and confirmed root causes to continuously evaluate and tune Vertex AI models.



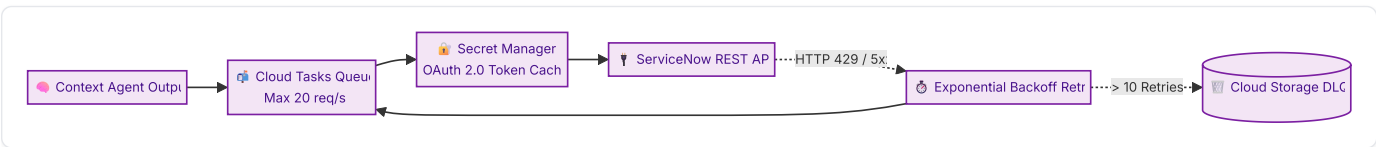
Syntax error in text
mermaid version 10.9.8

2. Outbound Dispatch Architecture & API Resiliency

During severe infrastructure disruptions or peak retail events (e.g., Black Friday / Cyber Monday), hundreds of secondary alerts can fire simultaneously. Direct synchronous REST API calls risk **HTTP 429 (Too Many Requests)** throttling by ServiceNow instances.

2.1 Asynchronous Rate-Limited Dispatcher (Cloud Tasks)

- **Queue Configuration:** A dedicated **Cloud Tasks Queue** (`aiops-snow-dispatch-queue`) acts as a shock absorber.
- **Dispatch Rate:** Capped at `20\text{ requests/second}` (configurable per enterprise ServiceNow license tier).
- **Exponential Backoff:** On HTTP 429, 500, or 503 responses, Cloud Tasks retries exponentially (`1\text{ s}`, `2\text{ s}`, `4\text{ s}`, `8\text{ s}`, `\dots`, `60\text{ s}`) up to 10 attempts before routing to the dead-letter topic `aiops.snow.dlq`.



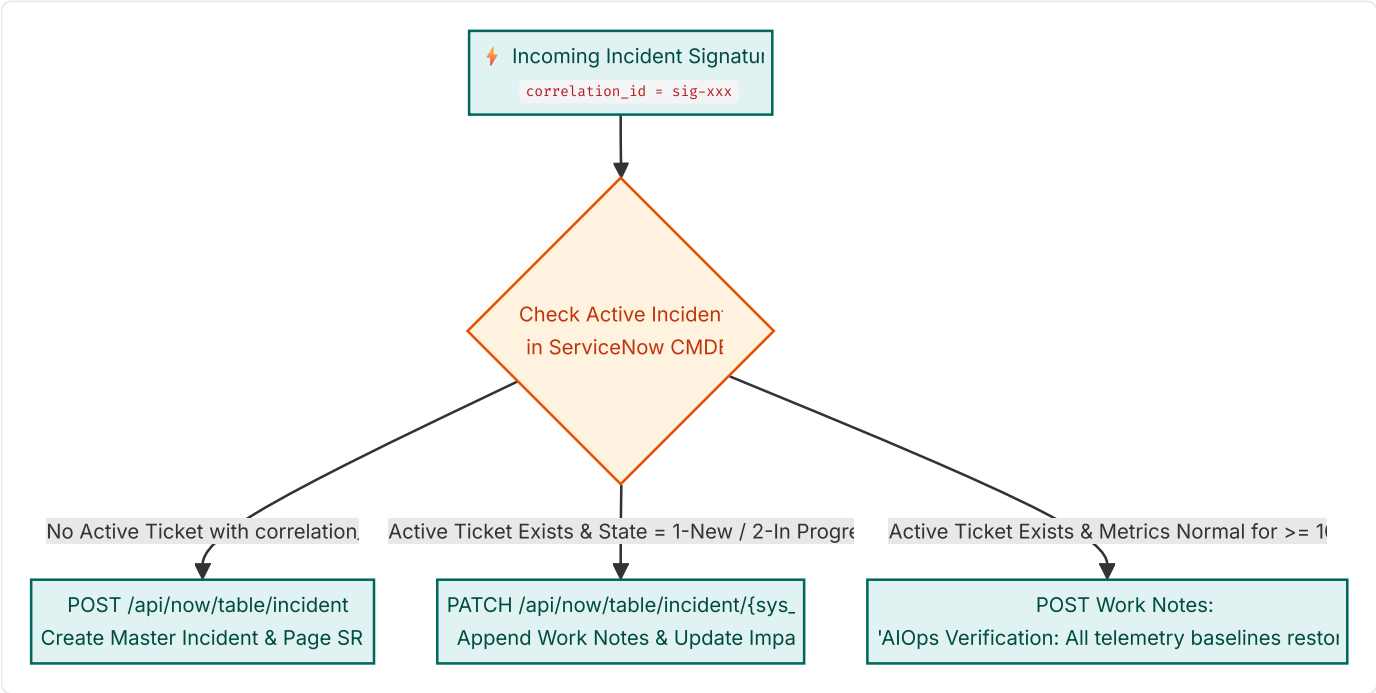
2.2 Enriched Incident Creation Payload Contract

- **Protocol:** HTTPS REST API (`/api/now/table/incident`).
- **Authentication:** OAuth 2.0 Client Credentials with cached bearer tokens.

```
{
  "short_description": "P1 Anomaly: Checkout API Latency Surge causing $14.2k/min revenue drop",
  "urgency": "1",
  "impact": "1",
  "assignment_group": "E-Commerce SRE Pod",
  "cmdb_ci": "ci_sys_id_checkout_api",
  "correlation_id": "sig-2026-08-26-chk-001",
  "correlation_display": "GCP AIOps Engine",
  "work_notes": "### 📄 AIOps Context-Aware RCA Summary\n- **Root Cause:** Database Connection Pool Exhausted on node `checkout-db-primary` (Dynatrace"
}
```

3. Incident Lifecycle, Deduplication & State Synchronization

To avoid duplicate ticket spam when cascading alerts continue to arrive during an ongoing outage, the platform enforces strict lifecycle synchronization:



3.1 Deduplication & Update Rules

- 1. **Correlation Keying:** Every incident payload carries correlation_id = incident_signature_id (derived from the affected root CMDB CI and time window).
- 2. **Existing Incident Check:** Before creating a ticket, Cloud Tasks checks for existing records where correlation_id = signature_id and incident_state NOT IN (6, 7) (not Resolved or Closed).
- 3. **In-Flight Enrichment:** If an incident is already active, subsequent findings (e.g., additional Splunk logs or escalating financial impact) are appended to the existing ticket's work_notes with a PATCH request rather than opening a new ticket.
- 4. **Auto-Recovery Verification:** When telemetry baselines return to normal for >= 10 minutes, the Semantic Router automatically posts an Operational Recovery Note to the incident, prompting the on-call engineer to verify and resolve the ticket.

4. Multi-Cloud CMDB Topological Mapping

To route incidents accurately without human intervention, multi-cloud telemetry entities are mapped continuously to ServiceNow CMDB Configuration Items (CIs):

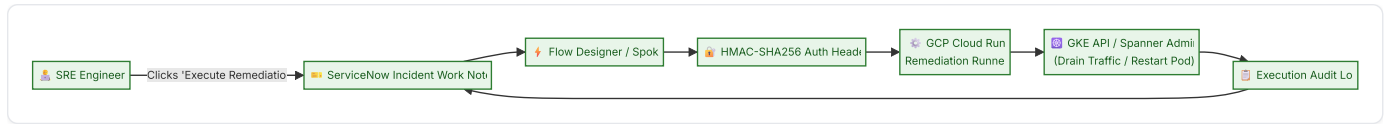
Telemetry Source	Entity Format / Identifier	ServiceNow CMDB Target Table	SRE Ownership Pod Resolution
GCP GKE	k8s.pod.name , k8s.namespace.name	cmdb_ci_kubernetes_pod / cmdb_ci_appl	namespace metadata → Pod Matrix
Dynatrace	Smartscape Entity ID (SERVICE-XXX , HOST-XXX)	cmdb_ci_service / cmdb_ci_server	Dynatrace Management Zone tags
Akamai	Edge Property / CDN Hostname (store.retail.com)	cmdb_ci_endpoint / cmdb_ci_web_server	Edge Networking Pod
Splunk	Hostname / POS Terminal ID (pos-lane-442)	cmdb_ci_vm_instance / cmdb_ci_pos	Store Systems Support Pod
Adobe Analytics	Funnel Domain (checkout , cart , search)	cmdb_ci_business_app	Digital Experience SRE Pod

4.1 Topology Synchronization Pipeline

- **Batch Sync:** A scheduled Cloud Composer DAG queries /api/now/table/cmdb_rel_ci nightly to refresh parent-child service dependency graphs in BigQuery (aiops_lakehouse.topology_service_graph).
- **Dynamic Resolution:** When an anomaly occurs on storefront-ui , the Semantic Router traverses the graph downward to identify if the root dependency is checkout-service or spanner-db , routing the ticket to the exact owner pod.

5. Human-in-the-Loop One-Click Automated Remediation

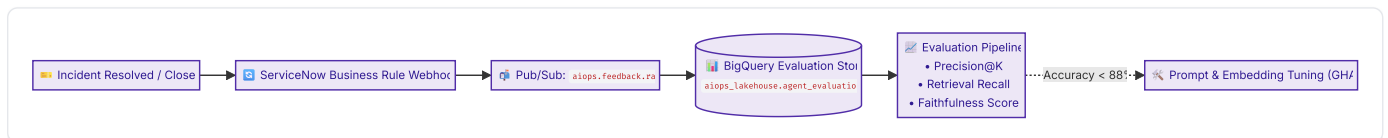
While diagnostics run autonomously, mutating remediation actions strictly enforce Human-in-the-Loop (HITL) governance:



- Remediation Trigger:** The Gemini SRE Agent attaches an approved runbook action block in ServiceNow work notes (e.g., `[Approve Traffic Drain for checkout-pod-x89]`).
- Execution Gate:** The on-call engineer clicks the approval button in the ServiceNow UI.
- Remediation Webhook:** **ServiceNow Flow Designer (Integration Hub)** sends a signed HTTPS POST request (`HMAC-SHA256`) to the **GCP Cloud Run Remediation Service**.
- Audit Logging:** The remediation runner executes the action (e.g., cordoning a degrading GKE node or scaling up database connection pools) and writes the complete terminal execution log back to ServiceNow `work_notes`.

6. Closed-Loop SRE Feedback & Continuous Model Learning

To continuously improve reasoning accuracy and avoid model hallucination drift, ServiceNow acts as the ground-truth data collection hub:



6.1 Feedback Webhook Payload Schema

When an engineer closes or resolves an incident, a ServiceNow **Business Rule** triggers an outbound REST webhook:

```

{
  "event_type": "incident_closed_feedback",
  "incident_id": "INC0948201",
  "correlation_id": "sig-2026-08-26-chk-001",
  "assigned_sre_pod": "Checkout-Backend-Pod",
  "ai_predicted_root_cause": "Spanner Connection Pool Saturation",
  "actual_confirmed_root_cause": "Spanner Connection Pool Saturation",
  "sre_accuracy_rating": 5,
  "false_positive_flag": false,
  "diagnostics_usefulness_rating": 4,
  "resolution_code": "Resolved by Runbook (Scaled Pool Replicas)",
  "duration_seconds": 680,
  "closed_at": "2026-08-26T06:41:20Z"
}
  
```

6.2 BigQuery Model Evaluation Store

Feedback payloads stream directly into `aiops_lakehouse.agent_evaluations`:

- Routing Accuracy Metric:** Target $> 92\%$ correct pod routing.
- SOP Retrieval Recall@3:** Target $> 88\%$ relevant runbook retrieval.
- Automated Alerting:** If the rolling 7-day accuracy drops below target thresholds, a GitHub Actions workflow is triggered to re-tune reasoning prompts and re-index vector embeddings.

7. Operational Impact & MTTD / MTTR Quantification

Incident Lifecycle Stage	Traditional Manual Operations	AIOps + ServiceNow Platform	Time Savings / MTTR Reduction
Detection (MTTD)	5 – 15 minutes (waiting for customer complaints or siloed alerts)	\$< 60\$ seconds (continuous BQML time-series & Davis AI webhooks)	\$85%\$ reduction
L1 Triage & Queue Routing	10 – 20 minutes (manual ticket dispatch by L1 helpdesk)	Instant (< 2 seconds) (Semantic Router direct pod routing)	\$100%\$ automation
SOP Runbook Retrieval	5 – 10 minutes (searching wikis and Confluence pages)	Instant (< 1 second) (Vertex AI Vector Search retrieves SOP)	\$100%\$ automation
Diagnostic Execution	10 – 25 minutes (running CLI queries across 5 tools)	Pre-executed (< 10 seconds) (findings attached before SRE arrival)	\$90%\$ reduction
Remediation Execution	15 – 30 minutes (manual command-line triage and execution)	1 – 3 minutes (one-click approved remediation runner)	\$80%\$ reduction
Total MTTR	45 – 100 minutes	8 – 15 minutes	\$> 75%\$ MTTR Improvement

8. Security, Access Controls & Auditability

1. **OAuth 2.0 Client Credentials:** All outbound requests to ServiceNow authenticate via scoped OAuth 2.0 tokens cached securely in Cloud Memorystore with secrets in **GCP Secret Manager**.
2. **Least Privilege ACLs:**
 - GCP Service Account: `create` and `update` permissions on `incident` and `sys_journal_field`; `read-only` on `cmdb_ci` and `cmdb_rel_ci`.
 - Webhook Ingress: Enforces IP allowlisting and HMAC-SHA256 signature verification.
3. **Audit Trail Transparency:** Every automated diagnostic query and remediation execution is stamped with full parameter transparency in ServiceNow Work Notes, fulfilling strict enterprise compliance and post-incident review (PIR) mandates.

9. Related Architectural Specifications

- [High-Level Platform Architecture](#)
- [Ingestion Master Blueprint](#)
- [Data Processing & AI Feature Store](#)
- [AIOps Intelligence Layer Specification](#)